

Module-2

RESISTIVE SENSORS

2.1 POTENTIOMETERS

A potentiometer is a resistive device with a linear or rotary sliding contact (Figure 2.1). The resistance between that contact and the bottom terminal is

$$R = \frac{\rho}{A}x = \frac{\rho l}{A}\alpha \quad \text{----- 2.1}$$

where ρ is the resistivity, A is the cross section, l is the length, x is the distance travelled from the bottom terminal, and α is the corresponding length fraction.

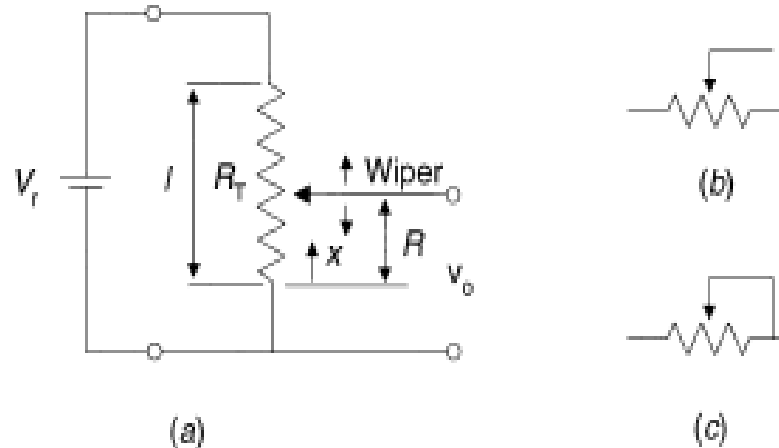


Figure 2.1 (a) Ideal (linear) potentiometer connected as a voltage divider and (b) its symbol; the arrow indicates that the resistance variation responds to a mechanical action. (c) The wiper can also be connected to an end terminal to obtain a variable resistor (rheostat).

A potentiometer is a zero-order system, although it can be itself a component of a nonzero-order sensor - for example, a mass-spring system. Equation (2.1) means that the resistance is proportional to the travel of the wiper. This implies the acceptance of several simplifications that may not necessarily be true. First, we assume that the resistance is uniform along the length l . But the resistance is not perfectly uniform, which limits the linearity of the potentiometer. The agreement between the actual and the theoretical transfer characteristic (here a straight line) is termed conformity (here linearity). Second, we assume that the sliding contact gives a smooth resistance variation, not a stepped one, and therefore that the resolution is infinite. But that is not true for wound resistive elements or for conductive-plastic potentiometers. For (2.1) to be valid, if the potentiometer is supplied by an alternating voltage, its inductance and capacitance should be insignificant. For low values of the total resistance R_T , the inductance may be significant, particularly in models with wound resistive elements. For high values of R_T , the parasitic capacitance may be important. Furthermore, resistors drift with temperature. Therefore (2.1) is valid only if resistance changes due to temperature are uniform. Temperature changes can arise not only from fluctuations in ambient temperature but also from self-heating due to the finite power that the potentiometer dissipates. If the power rating is P , the maximal rms value of the applied voltage V_r must be

$$V_r < \sqrt{PR_T} \quad \text{----- 2.2}$$

Measurement circuits with relatively low input impedance have an electrical loading effect on the potentiometer, and some parts of the potentiometer may heat beyond their rated power.

Friction and inertia of the wiper also limit the model validity because they add a mechanical load to the system being measured. These should be insignificant but at the same time ensure a good contact. As a compromise the force required to displace the wiper is from 3g to 15g.

For variable movements the starting torque is approximately twice the dynamic torque, and this is reduced by lubrication. For rapid movements there is a risk of losing contact during vibration. Thus, some units have two wipers of different arm lengths, and therefore different resonant frequencies. Alternatively, some models have elastomer damped wipers. Maximal travel speed is limited to about 10 m/s. The axis of rotary potentiometers must be concentric to the axis whose angular displacement is to be measured.

Finally, noise associated with the wiper contact limits resolution. When contact resistance changes with movement from one position to another, current circulating through changes the output voltage, and these fluctuations may be significant for the attached measuring device. Noise can increase because of dust, humidity, oxidation, and wear. Most of these limitations are outweighed by the advantages of this device. It is simple and robust and yields a high-level voltage with high accuracy relative to its cost. Models available accept linear and rotary movements (one or more turns in helical units). In some models the output is deliberately nonlinear with respect to the displacement. For example, the output can be a trigonometric function - sine, cosine, tangent - of the angle turned by the sliding contact. A nonlinear relationship can also be obtained by using a nonuniform spacing for the wire or by varying its size along its length. When the measuring circuit loads the potentiometer, the result is also a nonlinear characteristic.

Dual potentiometers operated by a single control stick (joysticks) move in four quadrants to locate a point in a plane. Movement along the x-axis controls R_x , and movement along the y-axis controls R_y (Figure 2.2). If both potentiometers are supplied by the same voltage, the output voltages are

$$\begin{aligned} v_x &= V_r(1 - 2\alpha) \\ v_y &= V_r(1 - 2\beta) \end{aligned} \quad \text{----- 2.3}$$

where $0 \leq \alpha, \beta \leq 1$. At the center of the plane we have $\alpha = \beta = 0.5$, and the voltages obtained are (0 V, 0 V).

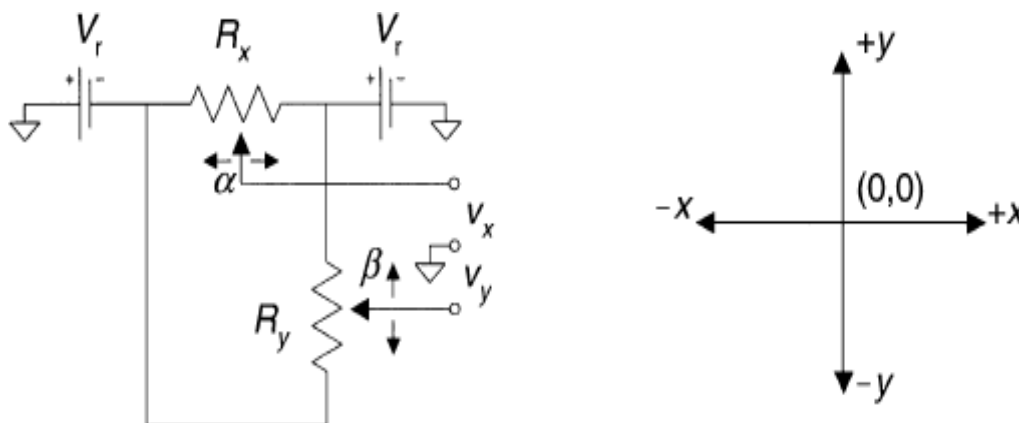


Figure 2.2 Joystick based on a dual potentiometer operated by a single control stick able to move around in four quadrants.

Potentiometers comprise a resistive element, a wiper, an actuating or driving rod, bearings, and housing. The resistive element is not a single wire. Even if the wire were very thin (while retaining enough strength), it would be impossible to obtain a high-enough resistance value as compared to connecting wires. One common configuration is a wire wound around a (ceramic)insulating form.

Some of the wire materials used are nickel-chrome, nickel-copper, and precious metal alloys. But then the inductance is high and the resolution is low. Advantages are a low temperature coefficient and a high-power rating.

Electrolytic or liquid potentiometers have a particular arrangement intended for tilt measurement. There is a hermetic, curved vial partially filled with a conductive liquid and three metal electrodes contacting the fluid, a large central electrode, and two shorter electrodes, one at each side (Figure 2.3a). A 0.5 V to 12.5 V ac voltage - 20 Hz to 20 kHz - is applied to the outer electrodes (a dc voltage would electrolyze the liquid). When the tube is horizontal, the electric resistance from the center electrode to each outer electrode is equal and the voltage at the central electrode is half the applied voltage (i.e., 0 V). As the tube tilts, the air pocket shifts, the current path between outer electrodes changes, and so does the resistance between the central electrode and each outer electrode. The output voltage is proportional to the tilt angle. The measurement span available in different models ranges from $\pm 0.5^\circ$ to $\pm 60^\circ$, and the resolution is up to 10^{-8} rad. Response time ranges from 100 ms to several minutes. Temperature interferes because it changes the resistivity of the electrolyte. Tilt switches use two close electrodes and mercury that closes the circuit when the vial rotates by a given angle.

Figure 2.3b shows another potentiometer intended for inclination measurement. A vertical planar substrate supports a concentric conductor and resistor. A conductive ball is pressed against the resistor by a channeled housing ring (not shown in the figure) and acts as wiper. When the housing tilts, the ball rolls to the lowest point and changes the resistance between the conductor and the resistor terminals. The range is about 360° and the resolution is 0.1° .

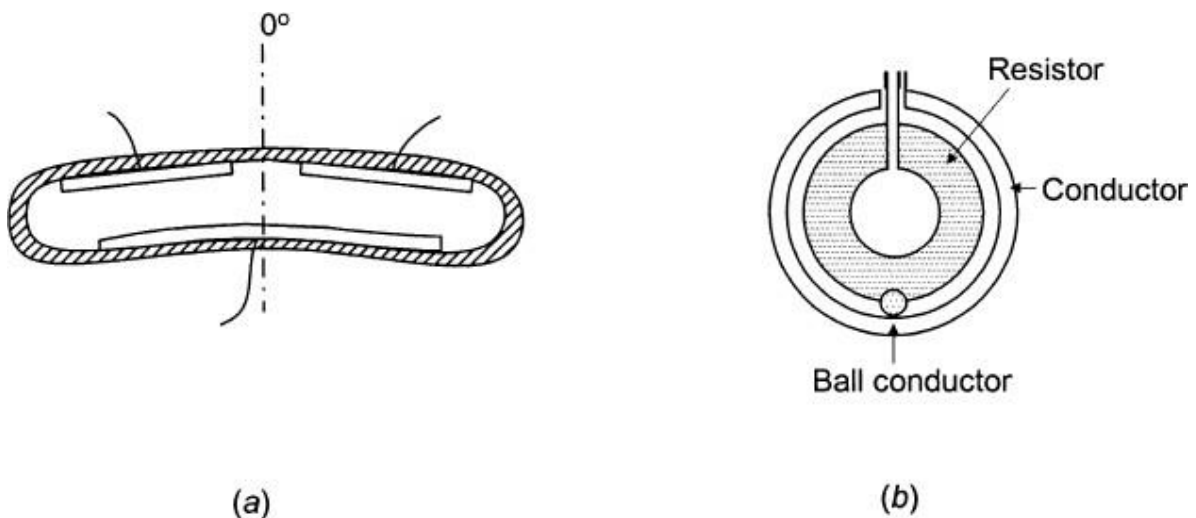


Figure 2.3 Tilt sensors based on potentiometers. (a) Liquid potentiometer. (b) A rolling ball acts as wiper in a pendulum, changing the resistance between the resistor ends and a conductor supported by a vertical plate.

The Thevenin equivalent circuit for a potentiometer shows that its output impedance depends on the wiper position. In a linear potentiometer supplied by a direct voltage, the output resistance R_o is the parallel combination of $R_T(1-\alpha)$ and $R_T\alpha$,

$$R_o = \frac{R_T\alpha R_T(1-\alpha)}{R_T\alpha + R_T(1-\alpha)} = R_T\alpha(1-\alpha) \quad \text{----- 2.4}$$

The open circuit output voltage is

$$v_o = V_r \frac{R_T\alpha}{R_T} = V_r\alpha \quad \text{----- 2.5}$$

where $0 \leq \alpha \leq 1$. The output voltage depends on both the supply voltage and the wiper position. The ratio between the output voltage and the supply voltage depends only on the wiper position. The output voltage is independent of R_T , but the output impedance increases with R_T .

2.2 STRAIN GAGES

2.2.1 Fundamentals: Piezoresistive Effect

Strain gages are based on the variation of resistance of a conductor or semiconductor when subjected to a mechanical stress. The electric resistance of a wire having length l , cross section A , and resistivity ρ is

$$R = \rho \frac{l}{A} \quad \text{----- 2.6}$$

When the wire is stressed longitudinally, each of the three quantities that affect R change and therefore R undergoes a change given by

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \frac{dl}{l} - \frac{dA}{A} \quad \text{----- 2.7}$$

The change in length that results when a force F is applied to a wire, within the elastic limit (Figure 2.4a), is given by Hooke's law,

$$\sigma = \frac{F}{A} = E\varepsilon = E \frac{dl}{l} \quad \text{----- 2.8}$$

where E is the Young's modulus (specific for each material and temperature-dependent), s is the mechanical stress, and e is the strain (unit deformation). e is dimensionless, but to improve clarity it is usually given in "microstrains" ($1 \text{ microstrain} = 1 \mu\varepsilon = 10^{-6} \text{ m/m}$). Strain in the elastic region is proportional to stress. The behavior in the plastic region is irreversible because there is strain after removing the applied force.

Consider a wire that in addition to a length l has a transverse dimension t (Figure 2.4b). A longitudinal stress changes both l and t . According to Poisson's law, we have

$$\nu = - \frac{dt/t}{dl/l} \quad \text{----- 2.9}$$

where ν is the Poisson ratio. The minus sign indicates that lengthening implies constriction. Usually, $0 < \nu < 0.5$, it is, for example, 0.17 for cast iron, 0.303 for steel, and 0.33 for aluminum and copper. Note that for the volume to remain constant, it should be $\nu = -0.5$, which is almost the case for rubber.

For a wire of circular cross section of diameter D , we have

$$A = \frac{\pi D^2}{4} \quad \text{----- 2.10}$$

$$\frac{dA}{A} = \frac{2dD}{D} = - \frac{2\nu dl}{l} \quad \text{----- 2.11}$$

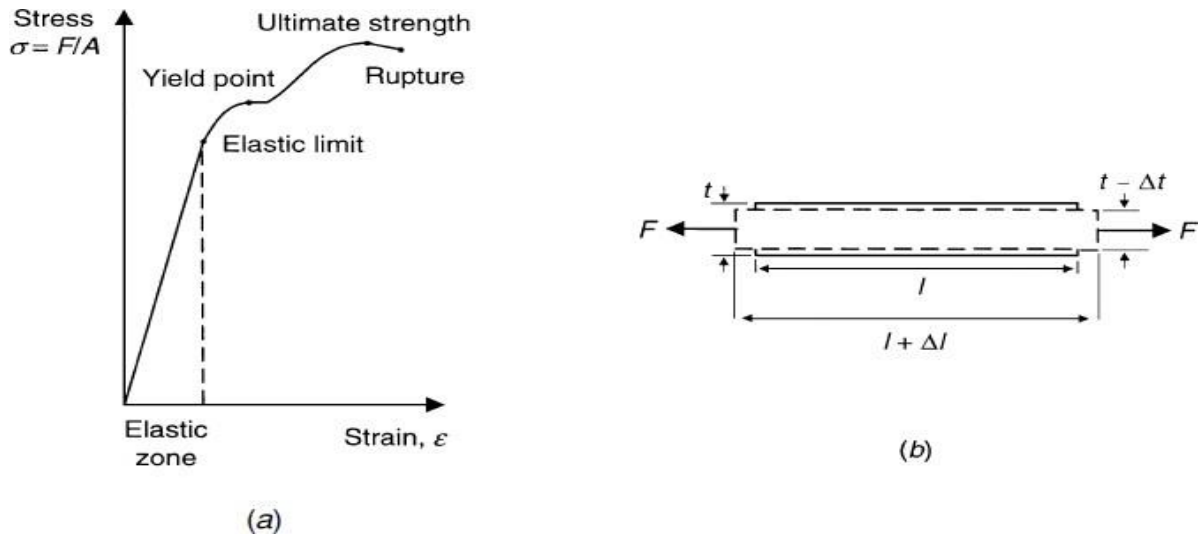


Figure 2.4 (a) Stress-strain diagram for mild steel. The elastic region has been greatly enlarged. (b) Action of Poisson's ratio: A longitudinal expansion implies a lateral contraction.

The change in resistivity as a result of a mechanical stress is called the piezoresistive effect. This effect results from the amplitude change of vibrations in the metal lattice. A longitudinal extension causes that amplitude to increase, which reduces electron mobility, thus increasing resistivity. P. W. Bridgman showed that, in metals, the percent changes of resistivity and volume are proportional:

$$\frac{d\rho}{\rho} = C \frac{dV}{V} \quad \text{----- 2.12}$$

where C is the Bridgman's constant. For the usual alloys from which strain gages are made, $1.13 < C < 1.15$. For platinum, $C = 4.4$. By applying (2.10), the change in volume can be expressed as

$$A = \frac{\pi D^2}{4} \quad \text{----- 2.13}$$

$$\frac{dA}{A} = \frac{2dD}{D} = -\frac{2\nu dl}{l} \quad \text{----- 2.14}$$

Therefore if the material is isotropic, within the elastic limit, (2.7) leads to

$$\frac{dR}{R} = \frac{dl}{l} [1 + 2\nu + C(1 - 2\nu)] = G \frac{dl}{l} = G\varepsilon \quad \text{----- 2.15}$$

where G is the gage factor, defined as the factor inside the square brackets. From the given values for ν and C, $G \approx 2$. For isoelastic, $G \approx 3.2$, for platinum, $G \approx 6$. Therefore for small variations, the resistance of the metallic wire is

$$R = R_0 + dR = R_0 \left(1 + \frac{dR}{R_0} \right) \cong R_0(1 + G\varepsilon) = R_0(1 + x) \quad \text{----- 2.16}$$

where R_0 is the resistance when there is no applied stress, and $x = G\varepsilon$. Usually, $x < 0.02$.

Example 1

A 350 Ω strain gage having $G = 2.1$ is attached to an aluminum strut ($E = 73$ GPa). The outside diameter of the strut is 50 mm and the inside diameter is 47.5 mm. Calculate the change in resistance when the strut supports a 1000 kg load.

From (2.15),

$$\Delta R = RG\varepsilon = RG \frac{F/A}{E}$$

From geometry, the area supporting the force is

$$A = \frac{\pi(D^2 - d^2)}{4} = \frac{\pi \times (97.5 \text{ mm}) \times (2.5 \text{ mm})}{4} = 191 \text{ mm}^2$$

Therefore, with $R = 350 \text{ } \Omega$, $G = 2.1$, $F = 1000 \text{ kg} = 9800 \text{ N}$, and $E = 73 \text{ GPa}$, we have

$$\Delta R = (350 \text{ } \Omega) \times 2.1 \times \frac{9800 \text{ N}}{(191 \times 10^{-6} \text{ m}^2) \times 73 \text{ GPa}} = 0.52 \text{ } \Omega$$

which is less than 0.15% of the initial resistance.

2.3 RESISTIVE TEMPERATURE DETECTORS (RTDs)

An RTD (resistance temperature detector) is a temperature detector based upon a variation in electric resistance. The commonest metal for this application is platinum, which is sometimes designated PRT (platinum resistance thermometer). Figure 2.5 shows the symbol for RTDs. The straight line diagonally crossing the resistor indicates that it changes linearly. The label near that line indicates that the change is induced by the temperature and has a positive coefficient. Three- and four-wire resistors reduce measurement errors from connecting leads.

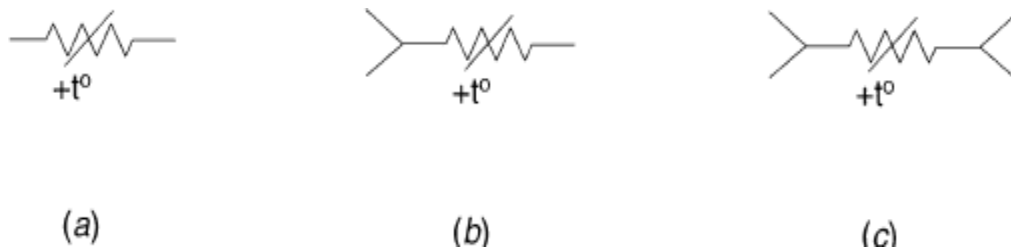


Figure 2.5 (a) Standard symbol for a resistor having a linear temperature dependence. Resistive sensors with three (b) and four (c) terminals permit lead compensation.

RTDs rely on the positive temperature coefficient for a conductor's resistance. In a conductor the number of electrons available to conduct electricity does not significantly change with temperature. But when the temperature increases, the vibrations of the atoms around their equilibrium positions increase in amplitude. This results in a greater dispersion of electrons, which reduces their average speed. Hence, the resistance increases when the temperature rises. This relationship can be written as

$$R = R_0[1 + \alpha_1(T - T_0) + \alpha_2(T - T_0)^2 + \dots + \alpha_n(T - T_0)^n] \text{----- 2.17}$$

where R_0 is the resistance at the reference temperature T_0 . The coefficients can be determined from resistance measurements at fixed-point temperatures: $0.01 \text{ } ^\circ\text{C}$ - (triple point of water), $100 \text{ } ^\circ\text{C}$ (boiling water), $660.323 \text{ } ^\circ\text{C}$ (freezing aluminum), and so on. The resistance changes because of both the change in resistivity and the change in dimensions caused by temperature.

Example 2

A given PRT has $100 \text{ } \Omega$ and $\delta = 6 \text{ mW/K}$ when immersed in air and $\delta = 100 \text{ mW/K}$ when immersed in still water. Calculate the maximal current through the sensor to keep the self-heating error below $0.1 \text{ } ^\circ\text{C}$.

The temperature increment above ambient temperature when dissipating a power P_D will be

$$\Delta T = \frac{P_D}{\delta} = \frac{I^2 R}{\delta}$$

Hence, the maximal current for a given temperature increment is

$$I = \sqrt{\frac{\Delta T \times \delta}{R}}$$

When the sensor is immersed in air, we obtain

$$I = \sqrt{\frac{(0.1^\circ\text{C})(0.006\text{ W/K})}{100\ \Omega}} = 2.4\text{ mA}$$

When the sensor is immersed in water, we have

$$I = \sqrt{\frac{(0.1^\circ\text{C})(0.1\text{ W/K})}{100\ \Omega}} = 10\text{ mA}$$

A dissipation factor more than 15 times higher in water than in air permits us to use a current just four times higher. Note that $1^\circ\text{C} = 1\text{ K}$, but temperatures expressed in degrees Celsius and kelvins are different.

Mechanical strain similar to that encountered by strain gages also limits temperature sensing by RTDs because it also changes the electric resistance. This interference may inadvertently arise when measuring surface temperatures with a bonded sensor. In surface measurement, temperature gradients may also cause errors. The principal advantages of RTDs are their high sensitivity (ten times that of thermocouples), high repeatability, long-term stability and accuracy for platinum ($0.1^\circ\text{C}/\text{year}$ in industrial probes, $0.0025^\circ\text{C}/\text{year}$ in laboratory probes), and the low cost for copper and nickel. RTDs use low-cost copper connections, an advantage compared to thermocouples. For metals used as RTD probes, in their respective linear range, (2.17) reduces to

$$R = R_0[1 + \alpha(T - T_0)] \quad \text{----- 2.18}$$

where α is the temperature coefficient of resistance (TCR), calculated from the resistance measured at two reference temperatures (e.g., 0°C and 100°C):

$$\alpha = \frac{R_{100} - R_0}{(100^\circ\text{C})R_0} \quad \text{----- 2.19}$$

α is sometimes termed relative sensitivity and depends on the reference temperature.

Example 3

A given PRT probe has $100\ \Omega$ and $\alpha = 0.00389\ (\Omega/\Omega)/\text{K}$ at 0°C . Calculate

- i) sensitivity
- ii) temperature coefficient at 25°C and 50°C .

The sensitivity is the slope of the resistance-temperature curve, here a straight line, hence with constant slope.

Sensitivity is

$$S = \alpha_0 R_0 = \alpha_{25} R_{25} = \alpha_{50} R_{50}$$

For the given sensor,

$$S = \frac{0.00389\ \Omega/\Omega}{\text{K}} \times (100\ \Omega) = 0.389\ \Omega/\text{K}$$

At 25 °C,

$$\alpha_{25} = \frac{\alpha_0 R_0}{R_{25}} = \frac{\alpha_0 R_0}{R_0 [1 + \alpha_0 (25^\circ\text{C} - 0^\circ\text{C})]} = \frac{\alpha_0}{1 + \alpha_0 (25^\circ\text{C})}$$

$$= \frac{0.00389(\Omega/\Omega)/\text{K}}{1 + (0.00389/\text{K}) \times (25^\circ\text{C})} = 0.00355(\Omega/\Omega)/\text{K}$$

At 50 °C,

$$\alpha_{50} = \frac{\alpha_0}{1 + \alpha_0 (50^\circ\text{C})} = \frac{0.00389(\Omega/\Omega)/\text{K}}{1 + (0.00389/\text{K}) \times (50^\circ\text{C})} = 0.00326(\Omega/\Omega)/\text{K}$$

Therefore, the temperature coefficient decreases for increasing temperature.

For immersion in fluids, there are models that consist of a thin wire noninductively wound around a ceramic form to allow some relative movement in order to make room for differential expansion (Figure 2.6a). An inert covering (stainless steel, glass), which may have one end of the resistance connected to it and grounded, protects the wire. Nevertheless, thin wires may break when vibrating. There are hand-held probes for surface temperature measurement, for which it is essential to be flexible and a good electrical insulator. Surface temperature sensing RTDs can also be mounted similarly to strain gages, and they can also be formed by parallel wire, foil, or deposited metallic film (Figure 2.6b).

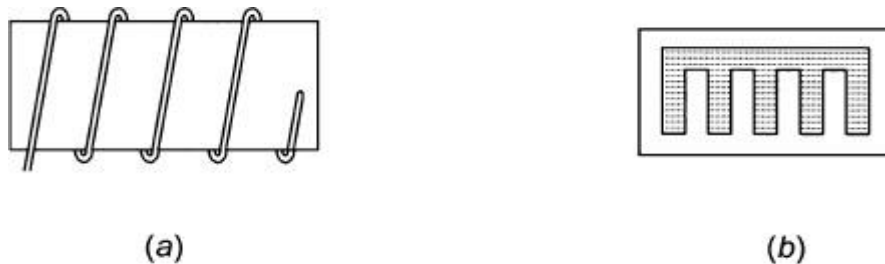


Figure 2.6 Platinum sensors for temperature probes use (a) wound or (b) film resistive elements.

The most common application for RTDs is temperature measurement. William Siemens first proposed platinum thermometers in 1871. Platinum probes offer a stable and accurate output, and for that reason they are used as calibration standards to interpolate between fixed-point temperatures in the International Practical Temperature Scale (IPTS). In those applications where platinum would be too expensive, nickel and its alloys are preferable. At very high temperatures, tungsten is used. They are extensively used to control thermal processes in the chemical industry, in automobiles (exhaust emission control, engine management), in domestic appliances (ovens), and buildings (central heating systems). In cars, for example, if the temperature of the catalytic converter decreases below 250 °C, it can become contaminated. Platinum temperature probes are also used to measure fluid velocity in the so-called hot wire anemometer. It is based on a very thin (25 μm in diameter) and short (0.2 mm to 25 mm) wire, held by its ends in a rigid support. An electric current passes through it in order to produce self-heating. When it is immersed in a fluid, the wire cools by convection, and consequently its electric resistance decreases. Obviously this probe would be useless if immersed in an electrically conductive fluid. Catalytic gas sensors use a fine coil of platinum wire and measure its temperature when heated at about 450 °C. The coil is embedded in a pellet (or “bead”) of sintered alumina powder, called a pellistor, impregnated by a catalyst (platinum, palladium, etc.) (Figure 2.7a).

If a flammable gas contacts the catalytic surface, it becomes oxidized (burns) - flameless - thus increasing the temperature of the wire by a few degrees, hence its resistance. In order to compensate for ambient temperature and humidity interference, a similar pellistor but without any catalyst is placed next to the active pellistor and series-connected to it (Figure 2.7b), thus forming a half-bridge. The pellistor couple is housed in a can open to the environment through a sinter disk. Flammable gases can penetrate the disk and reach the pellistors; but in case of ignition inside the sensor, the disk would cool the flame so that it would not flash back into the atmosphere. The sensitivity is higher and the response time is shorter for small molecules (hydrogen, methane, ammonia) than for larger molecules (octane, toluene, xylene) because small molecules diffuse better into the sensor, but otherwise cannot discriminate among similar gases. Pellistors are stable, reliable, and last for several years. However, sulfur, phosphorous, silicon, and lead compounds and corrosive substances poison the catalyst.

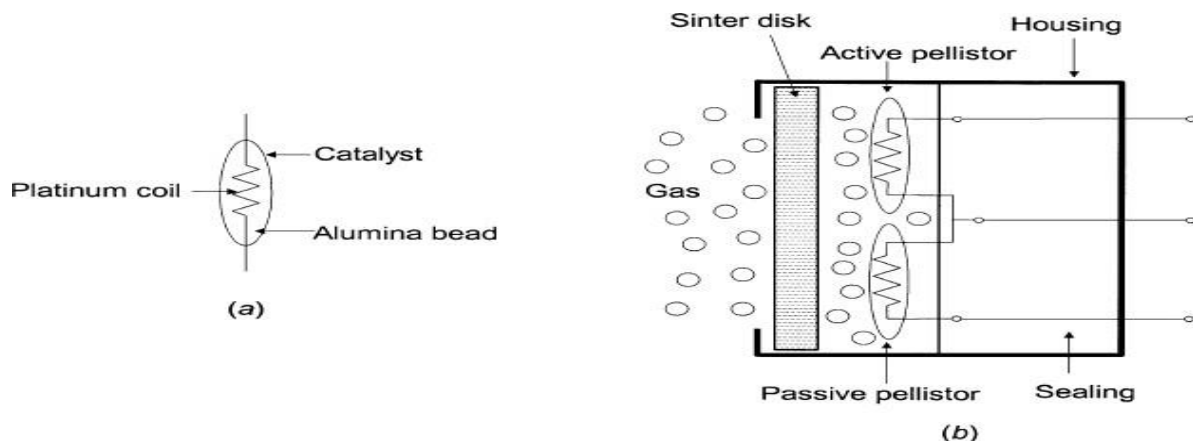


Figure 2.7 (a) Catalytic gas sensor based on a sintered bead with an embedded coil of platinum wire (pellistor) that (b) uses a passive pellistor connected to form a half-bridge for temperature and humidity compensation.

2.4 THERMISTORS

Models

Thermistor comes from “thermally sensitive resistor” and applies to temperature- dependent resistors that are based not on conductors as the RTD but on semiconductors. They are designated as NTC when having a negative temperature coefficient and as PTC when having a positive temperature coefficient. Figure 2.8 shows their respective symbols where the horizontal line at one end of the diagonal line indicates that the resistance variation is not linear. Michael Faraday described the first thermistor in 1833.

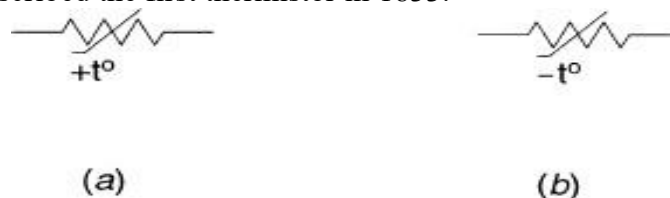


Figure 2.8 Standard symbol for a resistor having a nonlinear temperature dependence, with positive (a) or negative (b) TCR (IEC Publication 117-6).

Thermistors are based on the temperature dependence of a semiconductor's resistance, which is due to the variation in the number of available charge carriers and their mobility. When the temperature increases, the number of charge carriers increases too and the resistance decreases, thus yielding a negative temperature coefficient.

This dependence varies with the impurities; and when the doping is very heavy, the semiconductor achieves metallic properties and shows a positive temperature coefficient over a limited temperature range. For NTC thermistors, over a 50 °C span the dependence is almost exponential:

$$R_T = R_0 e^{B(1/T - 1/T_0)} \quad \text{----- 2.20}$$

where R_0 is the resistance at 25 °C or other reference temperature, and T_0 is this temperature in kelvins. For $R_0 = 25$ °C, $T_0 = 273.15 \text{ K} + 25 \text{ K} \approx 298 \text{ K}$.

Figure 2.9 shows actual R_T versus T curves for several units from different materials. B (or β) is called the characteristic temperature of the material; and its value, which is temperature-dependent, usually ranges from 2000 K to 4000 K. It increases with temperature.

The equivalent TCR, or relative sensitivity, is

$$\alpha = \frac{dR_T/dT}{R_T} = -\frac{B}{T^2} \quad \text{----- 2.21}$$

which shows a nonlinear dependence on T . At 25°C and taking $B = 4000 \text{ K}$, $\alpha = 4.5 \text{ %/K}$, which is more than ten times higher than that of the Pt100 probe. In general, the higher resistance units have higher TCR.

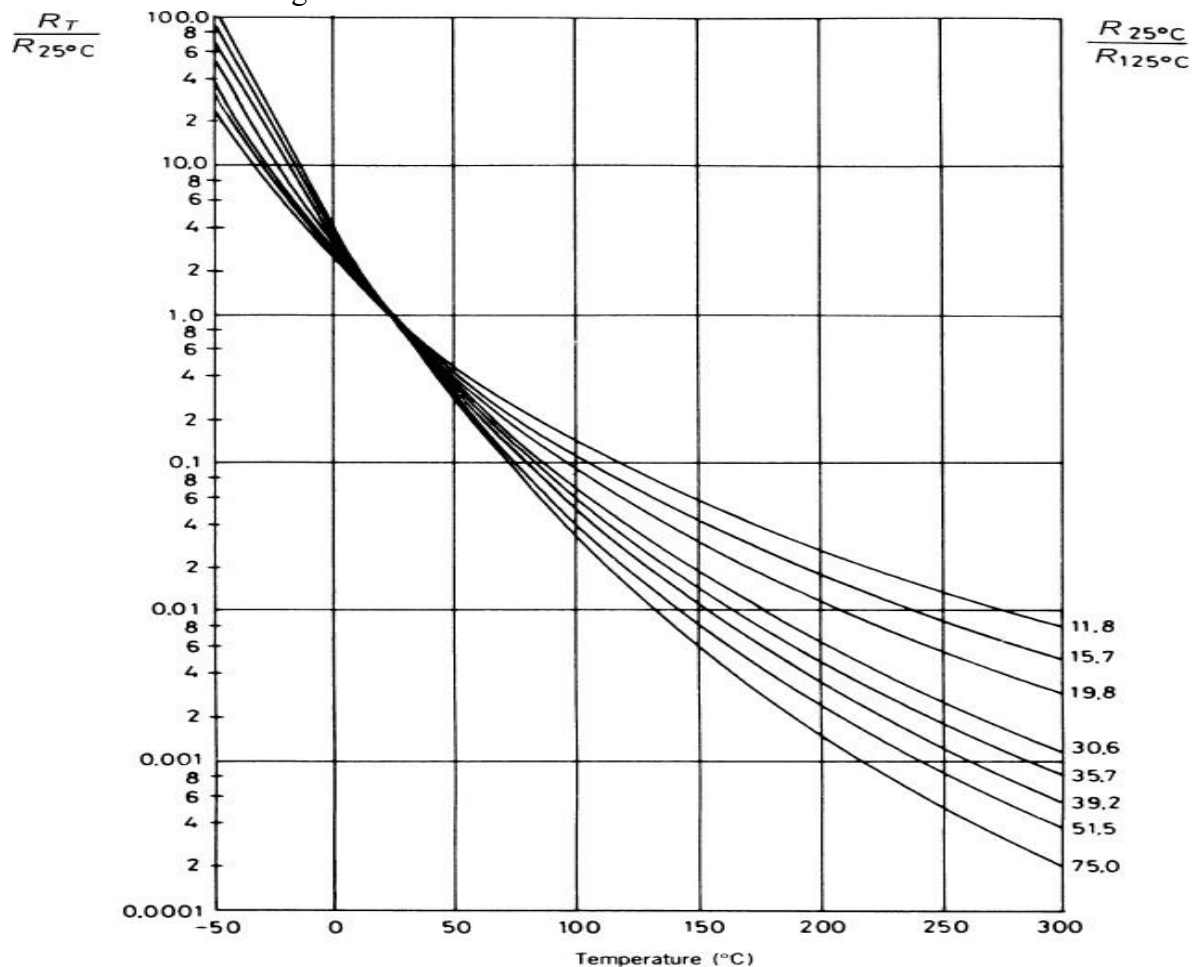


Figure 2.9 Resistance-temperature curve for several NTC thermistors (from Thermometrics).

Example 4

Given $R = Ae^{B/T}$. Determine A for a unit having $B = 4200 \text{ K}$ and $100 \text{ k}\Omega$ at 25 °C. Calculate the value for α at 0 °C and 50 °C.

$$A = R_0 e^{-B/T_0} = (100 \text{ k}\Omega) e^{(-4200 \text{ K})/(273.15 \text{ K} + 25 \text{ K})} = 0.0762 \text{ }\Omega$$

At 0 °C (= 273.15 K),

$$\alpha_0 = \frac{-4200 \text{ K}}{(273.15 \text{ K})^2} = -0.0563/\text{K}$$

At 50 °C (= 323.15 K),

$$\alpha_{50} = \frac{-4200 \text{ K}}{(323.15 \text{ K})^2} = -0.0402/\text{K}$$

B can be calculated from the NTC thermistor resistance at two reference temperatures T_1 and T_2 . If the measured resistances are, respectively, R_1 and R_2 , successively replacing these values in (2.20) and solving for B yields

$$B = \frac{\ln(R_2/R_1)}{\frac{1}{T_2} - \frac{1}{T_1}} \quad \text{----- 2.22}$$

Example 5

Calculate B for an NTC thermistor that has 5000 Ω at 25 °C and 1244 Ω at 60 °C.

From (2.22),

$$B = \frac{\ln \frac{1244 \text{ }\Omega}{5000 \text{ }\Omega}}{\frac{1}{(273.15 + 60) \text{ K}} - \frac{1}{(273.15 + 25) \text{ K}}} = 3948 \text{ K}$$

Note that the result does not depend on the reference temperature.

For a typical thermistor, a two-parameter model yields a ± 0.3 °C accuracy for a 50 °C span. A three-parameter model reduces the error to ± 0.01 °C in a 100 °C span. The model is then described by the empirical equation of Steinhart and Hart,

$$R_T = e^{(A+B/T+C/T^3)} \quad \text{----- 2.23}$$

or, alternatively, by

$$\frac{1}{T} = a + b \ln R_T + c(\ln R_T)^3 \quad \text{----- 2.24}$$

Measuring R_T at three known temperatures and solving the resulting equation system yields a, b, and c. From these parameters, the value for R_T at a given temperature T is

$$R_T = \exp \left(\sqrt[3]{-\frac{m}{2} + \sqrt{\frac{m^2}{4} + \frac{n^2}{27}}} + \sqrt[3]{\frac{m}{2} - \sqrt{\frac{m^2}{4} + \frac{n^2}{27}}} \right) \quad \text{----- 2.25}$$

where $m = (a - 1/T) / c$ and $n = b/c$.

2.5 RESONANT SENSORS

Sensors based on a resonant physical phenomenon yield an output frequency that depends on a measurand affecting the oscillation frequency. They require a frequency-counter in order to measure either the frequency or the oscillation period based on an accurate and stable clock. The choice of method depends on the desired resolution and also on the available time for the measurement. Resonant structures of single-crystal silicon are particularly suited to IC integration. Sensors use either harmonic oscillators or relaxation oscillators. Harmonic oscillators store energy that changes from one form of storage to another, for example from kinetic energy in a moving mass to potential energy in a stressed spring. In relaxation oscillators there is a single energy storage form, and the stored energy is periodically dissipated through some reset mechanism. Quartz clocks are accurate enough to derive a time base for most sensor applications, but they drift with time and temperature. Time drifts arise from structural changes due to defects in crystal lattices, mechanical stress from supporting elements (that decrease as time passes and change after thermal cycling), and mass changes because of absorption and desorption of contaminant gases inside the crystal package.

2.5.1 Sensors Based on Quartz Resonators

Quartz is piezoelectric and therefore an applied voltage stresses the crystal. If the voltage alternates at a proper rate, the crystal begins vibrating and yields a steady signal. L_1 is associated with the mass of the crystal, C_1 with its elasticity or mechanical compliance, and R_1 with its internal friction (resulting in heat dissipation) when oscillating. C_0 is the electrostatic capacitance of the crystal between the electrodes plus the holder and the leads. For a crystal used in a 32.768 kHz oscillator, for example, $L_1 = 4451$ H, $C_1 = 5$ fF, $R_1 = 11.2$ kW, and $C_0 = 1.84$ pF. The presence of a resonant circuit in Figure 2.10a permits the crystal to be used in an oscillator. At series resonance

$$f_s = \frac{1}{2\pi\sqrt{L_1 C_1}} \quad \text{----- 2.26}$$

the crystal's reactive elements cancel and provide an effective impedance consisting only of R_1 (Figure 2.10b). As the frequency increases, the crystal behaves as a positive reactance in series with a resistance. At the antiresonant-frequency f_a , the crystal's reactance is maximal. The range from f_s to f_a is referred to as the crystal's bandwidth.

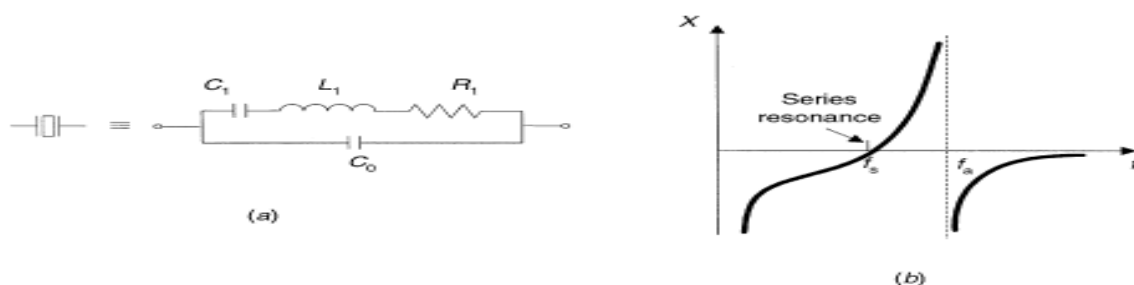


Figure 2.10 (a) High-frequency equivalent circuit for a piezoelectric material such as quartz with metal electrodes deposited on two faces. (b) The reactance of a quartz crystal varies with the operating frequency near resonance.

The series resonant circuit (Figure 2.11a) operates above f_s , where the reactance is slightly inductive. Series capacitance is then added to tune the circuit. Figure 2.11b shows a basic oscillator based on CMOS inverters. CL_1 and CL_2 are loading capacitances (larger for low oscillating frequency than for high oscillating frequency), $R_f = 1$ MW, and R_d is a damping resistor that depends on the gate type and oscillation frequency. Because quartz is inert, using a high-purity single crystal yields a mechanical resonance with large long-term stability.

Short-term stability depends on the quality factor Q (stiffness and low hysteresis) and equivalent inductance, which are very large. Short-term stability permits the design of high-resolution sensors. Long-term stability implies a longer time interval between calibrations.

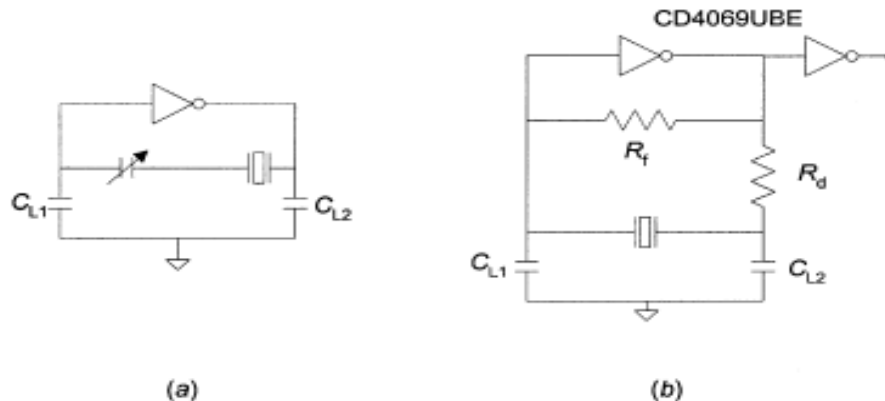


Figure 2.11 (a) Series resonator oscillator based on a quartz crystal. (b) Crystal (series) oscillator based on CMOS inverters.

2.5.1.1 Digital Quartz Thermometers

The values for the elements in the equivalent circuit for a quartz crystal depend on the temperature, and therefore the oscillation frequency displays a thermal drift. If precision-cut quartz crystals are used, the relationship between temperature and frequency is very stable and repeatable. Then, from the measurement of the oscillation frequency we can infer the temperature of the element. The general equation is

$$f = f_0[1 + a(T - T_0) + b(T - T_0)^2 + c(T - T_0)^3] \quad \text{-----} \quad 2.27$$

where T_0 is an arbitrary reference temperature (usually 25 °C), and f_0 , a , b , and c depend on the cutting orientation. Ideally, we would seek $b = c = 0$, but this is not easy. An alternative is to seek high sensitivity and repeatability instead of linearity and obtain T from $f - f_0$ by a look-up table (Figure 2.12). Some temperature probes using this principle include electronic circuitry to output a pulse frequency signal enabling remote sensing with low interference susceptibility as compared to systems with analog voltage output.

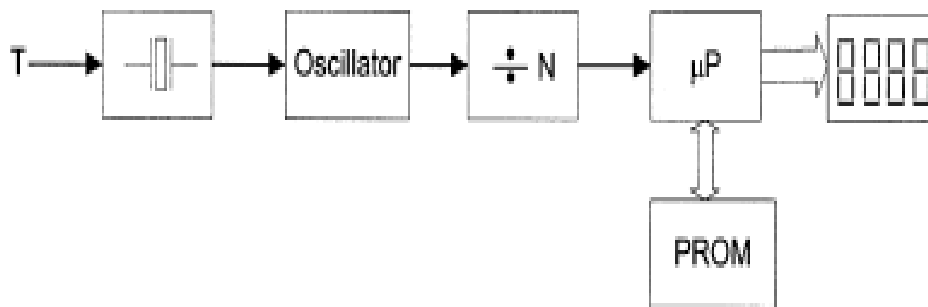


Figure 2.12 Simplified diagram for a quartz digital thermometer.

Oscillator frequencies range from about 256 kHz to 28 MHz with temperature coefficients (a) from $19 \times 10^{-6}/^{\circ}\text{C}$ to $90 \times 10^{-6}/^{\circ}\text{C}$. Sensitivities are up to about 1000 Hz/ $^{\circ}\text{C}$ in the range from -50°C to 150°C . The resolution can be as high as 0.0001°C , but the better the resolution, the slower the measurement speed. Some probes reach -40°C to 300°C , but with reduced linearity unless corrected by a look-up table. Low-mass probes can be applied to infrared radiation intensity measurement.

2.6 DIRECT SENSOR-MICROCONTROLLER INTERFACING

Interfacing sensors to microcontrollers without an intervening ADC requires a method to measure frequency, period, or time interval, which encodes the information about the measurand. Signals to be interfaced to microcontrollers must have the appropriate logical levels and fast voltage transitions (edges) to prevent false triggering by interference and metastable conditions.

2.6.1 Frequency Measurement

Frequency is usually measured by counting cycles during a known time interval. Figure 2.13 shows that this interval (gate time) is usually obtained from a precision clock by frequency dividers. The result is

$$N = f_x \times T_0 \quad \text{----- 2.28}$$

Because the input signal and the internal clock are asynchronous, we may stop counting just before the next input transition arrived or just after it arrived. This implies an uncertainty of 1 count, usually described by saying that the actual result is $N \pm 1$ counts. Because the resolution of this measuring method is 1 count, the relative resolution is $1/N$, which improves for large N . However, a large N implies a long measurement time, particularly for low frequencies. For example, measuring 10 kHz with uncertainty below 0.1% requires $N = 1000$ and, because each input cycle lasts 100 μ s, the measurement time will be 100 ms.

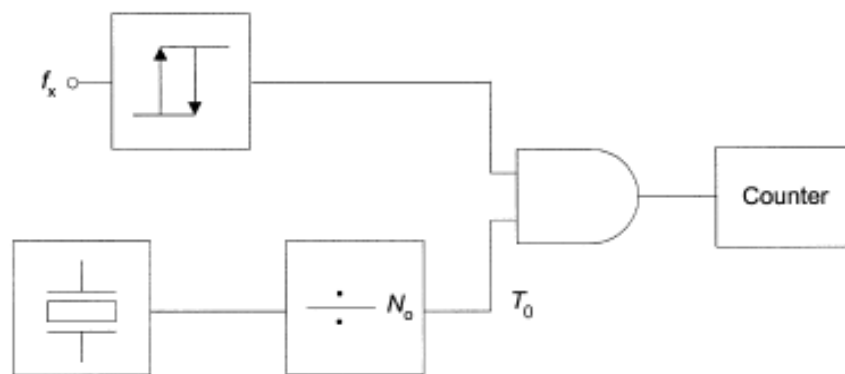


Figure 2.13 Basic block diagram of a frequency meter.

Example 1

A common sensor for traffic control is a buried flat coil supplied by a current from 20 kHz to 150 kHz. As a car enters the coil, its inductance decreases because of eddy currents induced in the car, and when the car leaves the coil the inductance returns to its original value. If the coil is included in a harmonic oscillator, the change in oscillation frequency signals the presence of a car. If the oscillation frequency changes by 10% because of a car and we measure it with an 8 bit counter, determine the maximal measurement time before the counter overflows.

The oscillation frequency of an LC harmonic oscillator is

$$f_0 = \frac{K}{\sqrt{LC}}$$

Therefore, if only the inductance changes

$$\frac{df_0}{f_0} = \frac{1}{2} \frac{dL}{L}$$

For a 10% change in inductance we can approximate

$$\Delta f_0 = -0.5 f_0 \frac{\Delta L}{L} = -0.5 f_0 (-0.1) = 0.05 f_0$$

The maximal frequency will therefore be

$$f_{\max} = f_0(1 + 0.05) = 1.05 \times 20 \text{ kHz} = 21 \text{ kHz}$$

Because the maximal reading of an 8 bit counter is 255, the counting time is limited to

$$T \leq \frac{255}{21 \text{ kHz}} = 12 \text{ ms}$$

Microcontrollers do not include a frequency divider able to provide the time base for frequency measurements according to Figure 2.16. An alternative method is to use two programmable counters, one to count the elapsed time and one to count the input pulses. For example, the 8051 includes two 16 bit timers/counters. When working as timers, the register is automatically incremented every machine cycle - which consists of 12 clock cycles for this particular model. When working as counters, the register is incremented in response to a 1-to-0 transition at its corresponding external input pin. Because recognizing such a transition takes two machine cycles, the maximal count rate is 1/24 the clock frequency. To measure frequency, one timer is preset at the measurement time T_0 and the input pulses are counted until the timer arrives at 0, at which moment it generates an interrupt that stops counting. Then the counter is read and the timer is reloaded at the preset time. If the frequency range goes from $f_{\min}$ to $f_{\max}$, the time needed to obtain a resolution of n bits ($N = 2^n$) is

$$T_0 = \frac{2^n}{f_{\max} - f_{\min}} \quad \text{----- 2.29}$$

For a given resolution, the narrower the frequency range, the larger the value of T_0 .

Example 2

A given sensor has an output frequency of 9 kHz to 11 kHz. Determine the measurement time and the number of counts needed to measure the frequency with a 12-bit resolution.

From (2.29),

$$T_0 = \frac{2^{12}}{11 \text{ kHz} - 9 \text{ kHz}} = 2.048 \text{ s}$$

A 9 kHz input yields $N_{\min} = (9 \times 10^3) \times (2.048 \text{ s}) = 18,342$.

An 11 kHz input yields $N_{\max} = (11 \times 10^3) \times (2.048 \text{ s}) = 22,528$.

Note that $N_{\max} - N_{\min} = 4096 = 2^{12}$ as desired, but each counter must have more than 14 bits.

2.6.2 Period and Time-Interval Measurement

Achieving high resolution when measuring low frequencies or narrow frequency ranges, which are common in sensors, takes a long time. This restricts dynamic measurements because the frequency signal must be constant while it is being counted. Measuring the signal period instead is faster, even when including the time needed to calculate the frequency from the measured period.

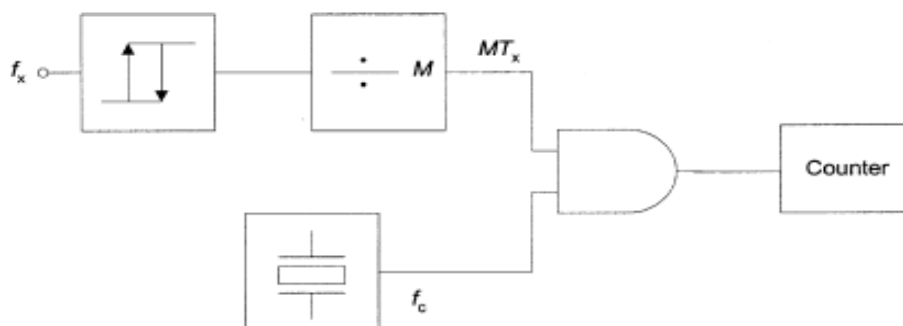


Figure 2.14 Basic block diagram of a period meter.

The period of a signal can be measured by counting pulses of a reference signal (clock) over a number of M periods (Figure 2.14). The result is

$$N = f_c M T_x \text{ ----- 2.30}$$

The measurement takes $M T_x$ and, because the resolution is $1/N$, from (2.30) we infer that the resolution-measurement time product is constant ($1/f_c$). For example, because a 10 kHz signal has 100 μ s period, measuring it with uncertainty below 0.1% requires $N = 1000$. If $f_c = 1$ MHz, we need $M T_x \cdot 1$ ms; that is, $M \cdot 10$. The input signal must be stable for 10 periods, far less than the 1000 periods needed to measure frequency with the same resolution. A faster clock would permit a shorter measurement time for the same resolution. Some microcontrollers cannot directly implement the method in Figure 2.17. Instead they can measure the total elapsed time over k input pulses. The result is $k T_x(m)$ where $T_x(m)$ is the period of the input signal in machine cycles. That is,

$$T_{x(m)} = \frac{1/f_x}{T_m} = \frac{f_c}{p f_x} \text{ ----- 2.31}$$

For the 8051, $p = 12$. To implement the measurement, one timer is used to measure the elapsed time $k T_x(m)$. (i.e., counting to k), and the sensor signal is connected to an external interrupt pin. Each 1 to 0 transition generates an interrupt. The interrupt routine counts sensor pulses; and when it gets to the predetermined k , it reads and clears the timer. k , which must be an integer, depends on the desired resolution. To obtain m bit resolution we need

$$k \geq \frac{2^m}{T_{\max(m)} - T_{\min(m)}} \text{ ----- 2.32}$$

where $T_{\max(m)}$ and $T_{\min(m)}$ are measured in machine cycles.

Example 3

A given sensor has an output frequency of 9 kHz to 11 kHz. Determine how many signal cycles must elapse to measure its period with 12 bit resolution by using an 8051 microcontroller whose clock frequency is 12 MHz.

In order to apply (2.32) we must first determine the maximal and minimal signal periods in machine cycles. From (2.31), with $p = 12$,

$$T_{\max(m)} = \frac{12 \text{ MHz}}{12 \times (9 \text{ kHz})} = 111.1 \text{ machine cycles}$$

$$T_{\min(m)} = \frac{12 \text{ MHz}}{12 \times (11 \text{ kHz})} = 90.9 \text{ machine cycles}$$

Applying now (2.32),

$$k = \frac{2^{12}}{111.1 - 90.9} = 202.8$$

Therefore, we should measure for 203 signal cycles.

If the information about the measurand is encoded in the frequency rather than in the period, the result is

$$f_x = \frac{k}{k T_{x(m)}} \text{ ----- 2.33}$$

Hence, in order to obtain an n bit resolution in the frequency we need to know the resolution required when measuring the time $k T_x(m)$. By taking the derivative of (2.33), we obtain

$$df_x = -\frac{k}{(kT_{x(m)})^2} d(kT_{x(m)}) = -\frac{f_x^2}{k} d(kT_{x(m)}) \quad \text{----- 2.34}$$

which means that the resolution in f_x is (f_x^2/k) times the resolution in $kT_{x(m)}$.

Therefore, to have n bit resolution in the calculated frequency, we must measure the period with a number of bits m such that

$$\frac{f_{\max} - f_{\min}}{2^n} \geq \frac{f_{\max}^2}{k} \frac{k(1/f_{\min} - 1/f_{\max})}{2^m} \quad \text{----- 2.35}$$

which leads to

$$m \geq n + \frac{\ln(f_{\max}/f_{\min})}{\ln 2} \quad \text{----- 2.36}$$

From this equation and (2.35), the predetermined number of cycles during which the microcontroller must count is

$$k \geq 2^n p \frac{f_{\max}}{f_c} \frac{f_{\max}}{f_{\max} - f_{\min}} \quad \text{----- 2.37}$$

where f_c is the clock frequency.

Example 4

Consider the same sensor in Examples 4 and 5 whose output frequency is from 9 kHz to 11 kHz. How many signal cycles must elapse to determine its frequency with a 12 bit resolution from the measurement of its period using an 8051 microcontroller whose clock frequency is 12 MHz? What is the resolution needed in the period measurement? How long does the measurement take?

From (2.37), with $p = 12$ and $f_c = 12$ MHz, we have

$$k \geq 2^{12} \times 12 \times \frac{11 \text{ kHz}}{12 \text{ MHz}} \times \frac{11}{11 - 9} = 247.8 \approx 248$$

From (2.36) we obtain

$$m \geq 12 + \frac{\ln(11/9)}{\ln 2} = 12.289 \approx 13 \text{ bit}$$

The measurement will be longer when the signal period is short. Because the period of a 9 kHz signal is 0.11 ms, 248 cycles will last 27.5 ms. This is shorter than the 2.048 s needed to achieve the same resolution from a frequency measurement.

The overall measurement time should include the time needed to determine the frequency by either computing the reciprocal of the period measured or by searching a look-up table. The size of this table will depend on $f_{\max} - f_{\min}$ and on the number of bits needed to represent each frequency value. Alternatively, the look-up table may store the values of the measurand whose information is encoded in the frequency. Some microcontrollers include timers that are turned on by a 0-to-1 transition in an external interrupt pin and that are turned off by a 1-to-0 transition in the same pin. This last transition also generates an interrupt that permits the interrupt routine to read and reset the timer. This operation mode enables the direct measurement of pulse width - for example, when the sensor produces a PWM modulation.

Example 5

We wish to measure a temperature from $-20\text{ }^{\circ}\text{C}$ to $60\text{ }^{\circ}\text{C}$ with $0.1\text{ }^{\circ}\text{C}$ resolution using a linearized silicon PTC $2000\text{ }\Omega$ thermistor with $\alpha = 0.79\text{ } \%/ \text{K}$ at $25\text{ }^{\circ}\text{C}$. The interface is a microcontroller that measures the time interval needed to charge an external capacitor C through the sensor up to the transition level needed to generate an interrupt. Determine C when that level is half the supply voltage applied to the resistor and the effective clock frequency is 1 MHz .

The voltage across the capacitor will rise exponentially according to

$$v_C(t) = V_{CC}(1 - e^{-t/R_TC})$$

The time needed to reach $V_{CC}/2$ is

$$t_{th} = R_TC \ln 2 = 0.693 R_TC$$

The model for the sensor is

$$R_T = R_0[1 + \alpha(T - T_0)] = (2000\text{ }\Omega)[1 + (0.0079/\text{K})(T - 25\text{ }^{\circ}\text{C})]$$

Therefore, the sensor has $1289\text{ }\Omega$ at $-20\text{ }^{\circ}\text{C}$ and $2553\text{ }\Omega$ at $60\text{ }^{\circ}\text{C}$. The corresponding times to reach the threshold voltage are

$$t_{th}(-20) = 0.693(1289\text{ }\Omega)C$$

$$t_{th}(60) = 0.693(2553\text{ }\Omega)C$$

The time difference is

$$\Delta t_{th} = 0.693(1264\text{ }\Omega)C = (876\text{ }\Omega)C$$

The dynamic range needed is

$$\text{DR} = \frac{60\text{ }^{\circ}\text{C} - (-20\text{ }^{\circ}\text{C})}{0.1\text{ }^{\circ}\text{C}} = 800$$

Because the resolution when counting is 1 (count), we need

$$\Delta N = 800 = f_c \Delta t_{th} = (1\text{ MHz})(876\text{ }\Omega)C$$

From this, $C = 0.9\text{ }\mu\text{F}$. We would select $C = 1\text{ }\mu\text{F}$. The maximal number of counts obtained would be 1769 at $60\text{ }^{\circ}\text{C}$ and the minimal 893 at $-20\text{ }^{\circ}\text{C}$. Because encoding the larger number requires 12 bits, and we need to store 800 different values, a look-up table should have about 1.2 kbyte.